Model Questions on Discrete Mathematics and Graph Theory (MA203)

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Method of Induction

1- Prove that $n^3 - n$ is divisible by 3 for every positive integer n by using mathematical induction

2- Show that $n^2 > 2n + 1$ for $n \geq 3$ by Mathematical Induction

3- Prove by mathematical induction that $1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1) = \frac{n(n+1)(n+2)}{3}$ for all $n \in \mathbb{N}$

4- Prove by Mathematical induction that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each positive integer n.

Warshall's Algorithm

5- Using Warshall algorithm, find the transitive closure of the relation $R = \{(1,2), (2,3), (3,3)\}$ on the set $A = \{1,2,3\}$.

6- Find the transitive closure of the relation $R = \{(1,2), (2,1), (2,3), (3,4)\}$ on the set $A = \{1,2,3,4\}$ using Warshall's algorithm

7- Apply Warshall's algorithm to compute the transitive closure of the relation on A = {a, b, c, d} represented by the matrix $M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

8- Find the transitive closure of the relation $R = \{(2,1), (2,3), (3,1), (3,4), (4,1), (4,3)\}$ on set $A = \{1, 2, 3, 4\}$ using Warshall's algorithm

Solution of Homogeneous Equation by Generating Function, Solution of Non-homogeneous Equation,

9- Solve the recurrence relation $a_n - 6a_{n-1} + 9a_{n-2} = 0$, for all $n \geq 2$ by using generating function.

10- Solve the recurrence relation $a_{n+2} - a_{n+1} - 2a_n = n^2$

11- Solve the recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ for $n \geq 2$, with $a_0 = 1$, $a_1 = 6$ using generating function

12- Find the solution of the recurrence relation $a_n = 2a_{n-1} + n$ for $n \geq 1$, with $a_0 = 1$.

13- Solve $a_n = a_{n-1} + 2a_{n-2}, n \geq 2$ with initial conditions $a_{0=0}, a_1 = 1$.

14-How many solutions are there of $x + y + z = 17$ subject to the constraints $x \geq 1$,$y \geq 2$ and $z \geq 3$

15-How many integers between 1 and 500 (inclusive) are

(i) divisible be at least one of 3, 5, 7?

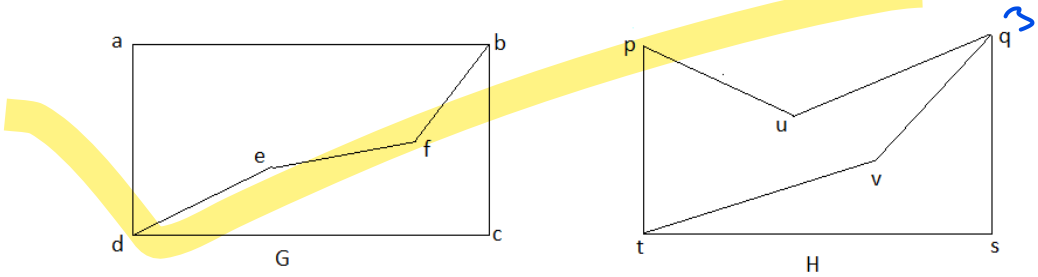
(ii) divisible by 3 and 5 but not by 7?

16-A committee of 5 members is to be formed from 6 men and 5 women. In how many ways can this be done when the committee consists of

(i) exactly 2 women, (ii) at least 2 women, (iii) at most 2 women

17-Prove that the number of vertices with odd degree is always even.

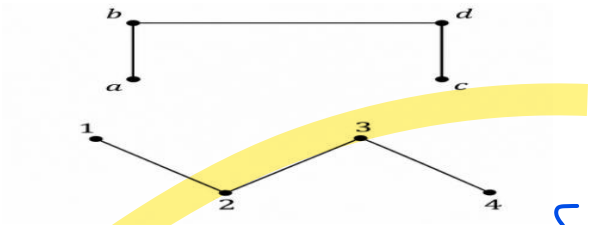
18-Test the isomorphism of the following graphs G and H:



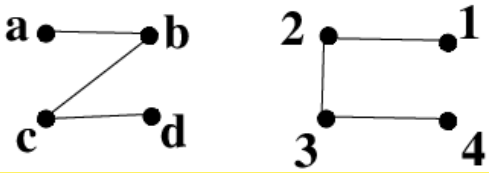
19-Show that Maximum number of edges in a simple graph with n vertices is $\frac{n(n-1)}{2}$

Test the isomorphism of the following two graphs

20-

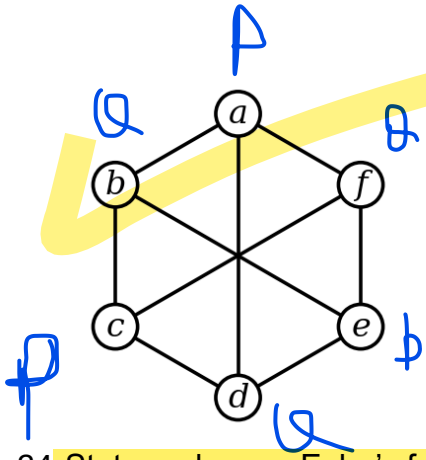


21-Show that the two graphs given below are isomorphic.



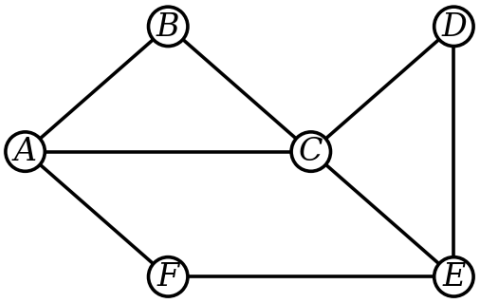
22-State and prove the Handshaking theorem. Hence show that there is no simple graph with degree sequence 3, 3, 2, 2, 1.

23-Define bipartite graph. Check whether the following graph is bipartite or not. If yes, write the bipartition.



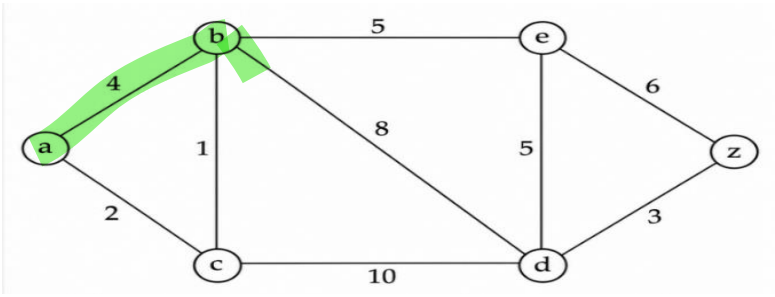
24-State and prove Euler's formula for connected planar graph

25-State the condition for a connected graph to have an Euler path. Use Fleury's algorithm to find a Euler path in the following graph.

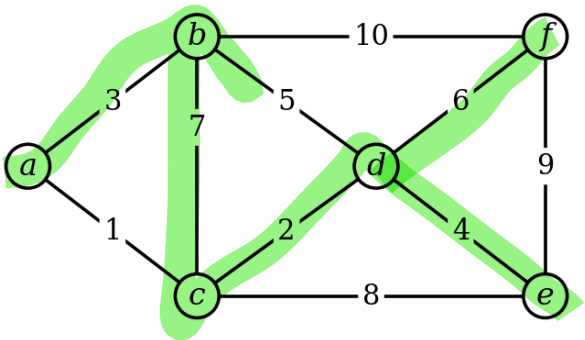


Prim's Algorithm for MST, Dijkstra's Algorithm, Ford-Fulkerson Algorithm.

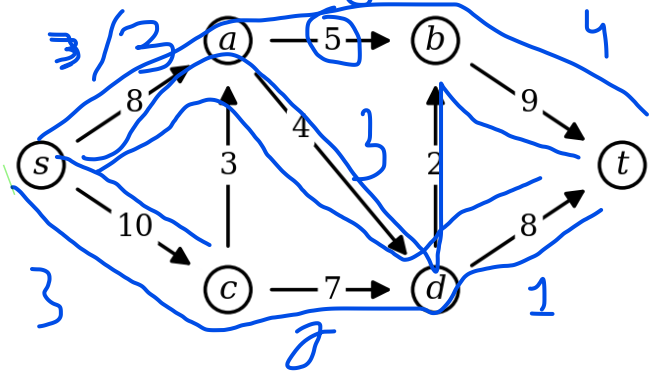
26-Find the length of a shortest path between the vertices a and z by using Dijkstra's algorithm in



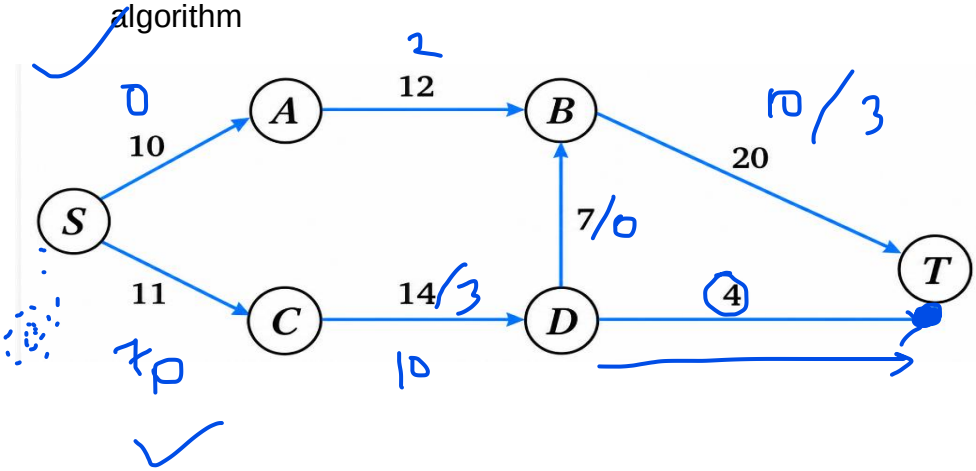
27-Use Prim's algorithm to find a minimal spanning tree (MST) of the following weighted graph, starting from vertex a. Also find its total weight.



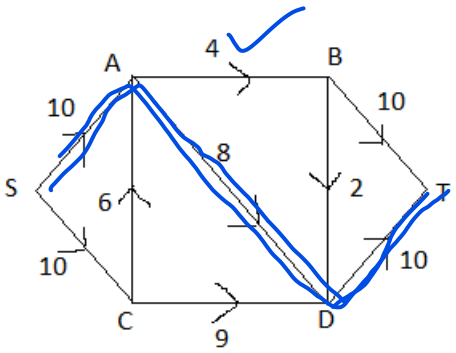
28-Find the maximum flow from s to t in the following network using the Ford-Fulkerson algorithm. Also write the corresponding minimum cut



29-Find the maximum flow of the following network using Ford-Fulkerson algorithm



30-Find the maximum flow of the following network using Ford-Fulkerson algorithm.



Checking whether an Algebraic Structure is a Group and Subgroup, Cosets, Group Codes, Error Detection and Correction.

31- Show that (2 ,5) encoding function defined by e(00)=00000, e(01)=10101, e(11)=11011 form a group code

32- Check whether the algebraic structure $G = (\left\{\begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in Q\right\}; *)$, where $A * B = A.B$ forms a group or not?

33- Let m= 2, n = 5, and

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Determine the group code $e_H: B^2 \rightarrow B^5$.

34- Let $G = (\mathbb{Z}_{12}, +)$ and $H = \{0, 4, 8\}$. Show that H is a subgroup of G. Find all the left cosets of H in G and verify Lagrange's theorem

35- Consider the encoding function $e: B^2 \rightarrow B^5$ defined by $e(00) = 00000, e(01) = 01110, e(10) = 10101, e(11) = 11011$. Show that e is a group code. Find its minimum distance and the number of errors it can detect and correct

36- Let $G = \{(a, b) | a, b \in R, a \neq 0\}$. Define a binary operation $*$ on G by $(a, b) * (c, d) = (ac, bc + d)$ for all $(a, b), (c, d) \in G$, show that $(G, *)$ is a group.

37- Let G be a finite group and H be a subgroup of G, then prove that the order of H divides the order of G